DATA FLOW DIAGRAM

& TRAFFIC MATRIX

Security Architecture Project

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1. Document Information

Document Title	Data Flow Diagram & Traffic Matrix
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1.1 Purpose

This document provides a comprehensive data flow diagram and traffic matrix for the organisation's network security architecture. It maps all authorised traffic flows between the 14 security zones defined in SEC-DOC-001 v2.0, specifying source, destination, protocol, port, direction, and the security control enforcing each flow.

This document serves as the primary reference for firewall rule validation, network access control reviews, and PCI-DSS data flow compliance evidence.

1.2 Scope

This document covers all inter-zone traffic flows across the 14 security zones confirmed in SEC-DOC-001 v2.0, including:

- Internet-facing flows through the Perimeter and DMZ zones
- Internal application flows across Application Services, Identity Services, and Core Network zones
- PCI-scoped flows through the ATM and Payment Processing zones
- SOC monitoring flows from all zones to the Log Collector and Wazuh SIEM
- Branch Office VPN connectivity flows
- Disaster Recovery replication flows from the Data Centre to the DR Site

1.3 Relationship to Other Documents

Reference	Document Title	Relationship
SEC-DOC-001	Security Architecture Overview v2.0	Parent document — zone definitions, IP inventory
SEC-DOC-002	Network Segmentation & VLAN Design	VLAN and subnet context for flows
SEC-DOC-003	Firewall Policy & Rules	Rule implementation of flows defined here
SEC-DOC-004	Log Collection & SIEM Architecture	Detail on monitoring flows in Section 6

2. Security Zone Trust Levels

All 14 security zones are classified by trust level. Traffic crossing trust boundaries must traverse an explicit security control. The table below summarises zone trust assignments used throughout this document.

#	Zone Name	Subnet	Trust Level	Key Control
1	Internet Zone	Public / Untrusted	UNTRUSTED	ISP Router / Border Router
2	Perimeter Security	203.0.113.x	LOW	Perimeter Firewall (203.0.113.4)
3	DMZ Network	172.16.10.x	LOW–MEDIUM	WAF / Reverse Proxy
4	DMZ Web Servers	172.16.10.20–30	MEDIUM	Internal Firewall
5	Internal Firewall Zone	172.16.21.x	MEDIUM–HIGH	Internal Firewall (172.16.21.10)
6	Application Services	172.16.20.x	TRUSTED	VLAN ACL / App Auth
7	Identity Services	172.16.60.x	TRUSTED	VLAN ACL / LDAPS
8	Core Network Infrastructure	10.0.0.x / 10.0.1.x	TRUSTED	Inter-VLAN ACL
9	ATM Network	Dedicated	ISOLATED / PCI	ATM Gateway (controlled exit)
10	Payment Processing	172.16.50.x	ISOLATED / PCI	Payment Firewall
11	SOC	172.16.66.x	HIGHLY TRUSTED	Receive-only monitoring paths
12	Data Centre	172.16.30.x	TRUSTED	DC Firewall / DB Firewall
13	Disaster Recovery Site	TBD	TRUSTED	Replication link from DC Backup
14	Branch Office	10.12.x.x	MEDIUM	IPSec VPN / Branch Firewall

3. Data Flow Descriptions

3.1 Flow 1 — Internet User to Banking Application

This flow describes the end-to-end path of an internet customer accessing the banking portal. The traffic traverses five distinct security layers before reaching the application tier.

Step	Source	Destination	Protocol / Port	Security Control
1	Internet User	ISP Router (203.0.113.1)	HTTPS / TCP 443	ISP-level DDoS filtering
2	ISP Router	Border Router (203.0.113.2)	HTTPS / TCP 443	WAN edge handoff
3	Border Router	DDoS Protection (in-line)	HTTPS / TCP 443	Volumetric scrubbing
4	DDoS Protection	Perimeter Firewall Outside (203.0.113.4)	HTTPS / TCP 443	Stateful inspection, geo-block
5	Perimeter FW DMZ (172.16.10.1)	WAF (172.16.10.10)	HTTPS / TCP 443	OWASP Top 10, SQLi, XSS block
6	WAF	Reverse Proxy (172.16.10.11)	HTTPS / TCP 443	Traffic normalisation
7	Reverse Proxy	Load Balancer (172.16.10.12)	HTTPS / TCP 443	Load distribution
8	Load Balancer	Web Server (172.16.10.20) / API Gateway (172.16.10.30)	HTTPS / TCP 443	Request serving
9	Perimeter FW Inside (172.16.20.1)	Internal Firewall (172.16.21.10)	HTTPS/API / TCP 443	Second-tier stateful inspection
10	Internal Firewall	Banking App (172.16.20.10) / Mobile Backend (172.16.20.11)	HTTPS / TCP 443	App-layer access control
11	Banking App	MFA Server (172.16.20.13) ↔ IAM Server (172.16.20.12)	HTTPS / TCP 443	MFA + IAM authentication
12	Banking App	Core Banking Server → DB Firewall → Core Banking DB (172.16.30.20)	TCP / SQL port	DB Firewall, least-privilege DB access

3.2 Flow 2 — Branch Office Connectivity

Branch office users access core internal resources via a site-to-site IPSec VPN tunnel terminated at the Core Routers.

Step	Source	Destination	Protocol / Port	Security Control
1	Branch Users	Branch Switch → Branch Firewall	All / internal	Branch Firewall inspection

Step	Source	Destination	Protocol / Port	Security Control
2	Branch Firewall	Branch Router (10.12.x.x)	All / internal	Branch perimeter control
3	Branch Router	Core Router 1 (10.0.0.1) / Core Router 2 (10.0.0.2)	IPSec / UDP 500 + 4500	Encrypted IPSec VPN tunnel
4	Core Routers	Core Switch 1 (10.0.1.1) / Core Switch 2 (10.0.1.2)	Routed / internal	Redundant dual-switch distribution
5	Core Switches	Authorised VLANs only (HR/Finance/IT/Admin)	VLAN-tagged / routed	Inter-VLAN ACL enforcement

3.3 Flow 3 — ATM and Payment Processing

ATM machines operate in a hard-isolated zone. The only authorised cross-zone path is from the ATM Gateway to the Payment Gateway. All payment processing is PCI-DSS scoped and isolated from the core network.

Step	Source	Destination	Protocol / Port	Security Control
1	ATM Machines	ATM Controller	ATM Protocol / dedicated	Isolated ATM network
2	ATM Controller	ATM Switch	ATM Protocol / dedicated	ATM zone switching
3	ATM Switch	ATM Gateway	ATM Protocol / dedicated	Controlled zone exit point
4	ATM Gateway	Payment Gateway (172.16.50.10)	HTTPS / TCP 443	Only authorised cross-zone link — Payment Firewall
5	Payment Gateway	Card Processing (172.16.50.20)	Internal / TCP	PCI-DSS isolated zone
6	Card Processing	Transaction Server (172.16.50.30)	Internal / TCP	PCI zone internal path
7	Transaction Server	Core Banking DB (172.16.30.20)	SQL / TCP	PCI-controlled path — DB Firewall

3.4 Flow 4 — DR Replication

Data Centre assets replicate to the Disaster Recovery site via confirmed dashed replication links. IP provisioning at the DR site is pending.

Step	Source	Destination	Protocol / Port	Security Control
1	Core Banking DB (172.16.30.20)	DR Replication Node (TBD)	Replication / TCP	Encrypted replication link
2	DR Replication Node	DR Database (TBD)	Internal / TCP	DR zone internal path
3	Backup (172.16.30.40)	DR Replication Node (TBD)	Replication / TCP	Encrypted backup replication link

Step	Source	Destination	Protocol / Port	Security Control
4	DR Replication Node	DR Backup (TBD)	Internal / TCP	Offsite backup storage

4. Complete Traffic Matrix

The following matrix documents all authorised inter-zone traffic flows. Any traffic not explicitly listed in this matrix is denied by default at the applicable security control boundary.

Source Zone	Destination Zone	Source Device	Destination Device	Port	Protocol	Security Control
Internet	Perimeter	User Browser	ISP Router	443	HTTPS	ISP DDoS Filter
Perimeter	DMZ	Perimeter Firewall	WAF (172.16.10.10)	443	HTTPS	Perimeter FW Stateful
DMZ	DMZ Web	Load Balancer	Web Server / API GW	443	HTTPS	DMZ Internal ACL
DMZ Web	App Services	Web Server / API GW	Internal FW (172.16.21.10)	443	HTTPS	Internal Firewall
App Services	App Services	Banking App	IAM / MFA Server	443	HTTPS	VLAN ACL (same /24)
App Services	Identity Svcs	Banking App	Active Directory	636	LDAPS	VLAN ACL / LDAPS
App Services	Identity Svcs	Banking App	RADIUS (172.16.60.12)	1812	RADIUS	VLAN ACL
App Services	Data Centre	Banking App	Core Banking Server	443	HTTPS	DC Firewall
Data Centre	Data Centre	Core Banking Server	Core Banking DB (172.16.30.20)	1433/3306	SQL	DB Firewall
Data Centre	Data Centre	Core Banking Server	Transaction DB (172.16.30.21)	1433/3306	SQL	DB Firewall
Data Centre	DR Site	Backup (172.16.30.40)	DR Replication Node	TCP	Replication	Encrypted Link
ATM Network	Payment PCI	ATM Gateway	Payment GW (172.16.50.10)	443	HTTPS	Payment Firewall
Payment PCI	Payment PCI	Payment GW	Card Processing	TCP	Internal	PCI Zone ACL
Payment PCI	Payment PCI	Card Processing	Transaction Server	TCP	Internal	PCI Zone ACL
Payment PCI	Data Centre	Transaction Server	Core Banking DB	SQL	TCP	DB Firewall / PCI Path
Branch Office	Core Network	Branch Router	Core Router 1 / 2	500, 4500	IPSec/UDP	IPSec VPN Tunnel
Core Network	App Services	Core Switch	Banking App	443	HTTPS	Inter-VLAN ACL

Source Zone	Destination Zone	Source Device	Destination Device	Port	Protocol	Security Control
Internet	DMZ	Email Sender	Email Gateway (172.16.10.41)	25/587	SMTP	Perimeter FW + Email Filter
Internet	DMZ	External Client	Public DNS (172.16.10.40)	53	DNS/UDP	Perimeter FW

5. Denied Traffic — Default-Deny Boundaries

The following traffic is explicitly blocked by the default-deny posture enforced at each security boundary. These denials are not exceptions — they represent the baseline position of the architecture.

Source Zone	Destination Zone	Blocked Traffic	Reason / Control
Internet	Application Services	Any direct access	No direct Internet → App path; must traverse Perimeter FW, DMZ, Internal FW
Internet	Data Centre	Any direct access	DC only reachable via Application Services; multiple firewalls in path
Internet	Identity Services	LDAP / AD queries	Identity Services zone is internal-only; no external DNS/LDAP exposure
ATM Network	Core Network	Any direct path	ATM zone is hard-isolated; only ATM Gateway → Payment GW link is authorised
Payment PCI	Core Network	Any non-DB path	Payment zone exits only via DB Firewall → Core Banking DB; no lateral access
DMZ	Data Centre	Any direct access	DMZ may not reach DC directly; must exit through Internal FW and App tier first
Branch Office	App Services (direct)	Any non-VPN path	Branch traffic is only permitted via IPSec VPN tunnel to Core Routers
App Services	Internet	Outbound direct	Application servers have no direct internet egress; all outbound routed via Proxy
VLAN HR (10)	VLAN Finance (20)	Any lateral traffic	Inter-VLAN ACLs block all cross-VLAN access not explicitly permitted

6. SOC Log Collection Data Flows

All zones transmit logs to the SOC Log Collector (172.16.66.20) via syslog (network devices) or Wazuh Agent (servers). Log data flows are receive-only — the SOC zone does not initiate connections back to source devices.

6.1 Syslog Flows — Network Devices

Source Device	Source Zone	Destination	Port	Protocol
Perimeter Firewall (203.0.113.4)	Perimeter	Log Collector (172.16.66.20)	514	Syslog / UDP
Internal Firewall (172.16.21.10)	Internal FW Zone	Log Collector (172.16.66.20)	514	Syslog / UDP
DC Firewall (172.16.30.1)	Data Centre	Log Collector (172.16.66.20)	514	Syslog / UDP
DB Firewall	Data Centre	Log Collector (172.16.66.20)	514	Syslog / UDP
Core Router 1 (10.0.0.1)	Core Network	Log Collector (172.16.66.20)	514	Syslog / UDP
Core Router 2 (10.0.0.2)	Core Network	Log Collector (172.16.66.20)	514	Syslog / UDP
Core Switch 1 (10.0.1.1)	Core Network	Log Collector (172.16.66.20)	514	Syslog / UDP
Core Switch 2 (10.0.1.2)	Core Network	Log Collector (172.16.66.20)	514	Syslog / UDP
Branch Firewall / Router	Branch Office	Log Collector (172.16.66.20)	514	Syslog / UDP
WAF (172.16.10.10)	DMZ Network	Log Collector (172.16.66.20)	514	Syslog / UDP
ATM Gateway	ATM Network	Log Collector (172.16.66.20)	514	Syslog / UDP

6.2 Wazuh Agent Flows — Servers

Source Device	Source Zone	Destination	Port	Protocol
Banking App (172.16.20.10)	App Services	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP
Mobile Backend (172.16.20.11)	App Services	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP
IAM Server (172.16.20.12)	App Services	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP
Active Directory (172.16.60.10)	Identity Services	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP
Core Banking Server	Data Centre	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP

Source Device	Source Zone	Destination	Port	Protocol
Core Banking DB (172.16.30.20)	Data Centre	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP
Transaction DB (172.16.30.21)	Data Centre	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP
Payment Gateway (172.16.50.10)	Payment PCI	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP
Transaction Server (172.16.50.30)	Payment PCI	Log Collector (172.16.66.20)	1514	Wazuh Agent / TCP

6.3 Internal SOC Flows

Source Device	Source Zone	Destination	Port	Protocol
Log Collector (172.16.66.20)	SOC	Wazuh SIEM (172.16.66.10)	TCP	Internal forward
Wazuh SIEM (172.16.66.10)	SOC	SOC Dashboard (172.16.66.30)	TCP	Dashboard feed

7. Revision History

Version	Date	Author	Changes
1.0	March 2026	Security Architecture Team	Initial release. Based on SEC-DOC-001 v2.0 fully resolved architecture. All 14 zones covered. Full traffic matrix, denied flows, and SOC log collection flows documented.

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